

SPATIAL TAXONOMY · PROOF OF CONCEPT

Building Width Calculator

Furniture → Zone → Width. A backwards-running method for sizing office buildings from tenant fit-out, driving the tool-buildingwidth engine.

AGENDA — FIVE SLIDES

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The Backwards Method

Most office buildings start with a width — sixty feet for daylight — then cram furniture in until it fits. Woodfine inverts the sequence: start with the tenant's chair, desk, exam table or bench, and the building width is whatever the arrangement requires. The cross-section has three zones, mirrored on each side of a central corridor.

Facade	Frontage	—
Zone 1	Habitat — desks	6 m
Zone 2	Magazine — storage	3.80 m
Zone 3	Corridor	2 m
Zone 2	Magazine — storage	3.80 m
Zone 1	Habitat — desks	6 m
Facade	Frontage	—
TOTAL	Professional Office	23.6 m 77'5"

Zone 1 — Habitat

Desks, workstations, exam tables, benches. Six metres minimum from the facade — the European Lighting Standard threshold for natural light at a workstation.

Zone 2 — Magazine

Storage, bookshelves, filing, server rooms. The slack variable: widen it to keep the plate from becoming a long, narrow rectangle.

Zone 3 — Corridor

Bilateral, mirrored at the centre. Width set by egress and accessibility — oversized by 20% for small-tenant traffic.

The 0.7 m German Circulation Law addition. Perpendicular desks at 6.0 m Habitat fit only two in series; 0.7 m unlocks the third with clearance. Token: `bim.circulation-addition.perpendicular-desk = 0.7 m`.

Zone Depth by Use Type

Each segment's width is the actual zone depth in metres, scaled against the widest total. The widest tenant (Civic) is roughly 1.4 m wider than the narrowest (Private Office) — the demising tolerance the floor plate must absorb.

Zone 1 — Habitat (desks)

Zone 2 — Magazine (storage)

Zone 3 — Corridor (circulation)

Private Office PO-1/PO-2/PO-3 — opens to shared building corridor	5.99 m	14.73 m 48'4" total
Academic Seating faces forward — Habitat under 6 m permitted	4.70 m 3 m	15.40 m 50'6" total
Professional Office Baseline professional office; tenant business suites	6 m 3.80 m 2 m	23.60 m 77'5" total
Laboratory Bench depth + fume hood clearance + IBC high-hazard egress	6.78 m 4.80 m 3.05 m	29.26 m 96'0" total
Medical Widest Habitat — exam table depth + patient circulation	7.20 m 4.87 m 2.89 m	29.92 m 98'2" total
Business Widest Magazine — open storage walls, server rooms	6 m 7.30 m 2.70 m	32.00 m 105'0" total
Civic Widest Corridor — public assembly egress	6 m 7.23 m 3.60 m	33.66 m 110'5" total

The Building Width Calculator

Zone 1 + Zone 2 + Zone 3 | Zone 3 + Zone 2 + Zone 1

Building Width = 2 × (Habitat + Magazine + Corridor)

USE TYPE	Z1 HABITAT	Z2 MAGAZINE	Z3 CORRIDOR	HALF	TOTAL WIDTH
Private Office	5.99 m / 19'8"	1.37 m / 4'6"	0.00 m / –	7.37 m	14.73 m / 48'4"
Academic	4.70 m / 15'5"	3 m / 9'10"	0.00 m / –	7.70 m	15.40 m / 50'6"
Professional Office	6 m / 19'8"	3.80 m / 12'6"	2 m / 6'7"	11.80 m	23.60 m / 77'5"
Laboratory	6.78 m / 22'3"	4.80 m / 15'9"	3.05 m / 10'0"	14.63 m	29.26 m / 96'0"
Medical	7.20 m / 23'7"	4.87 m / 16'0"	2.89 m / 9'6"	14.96 m	29.92 m / 98'2"
Business	6 m / 19'8"	7.30 m / 23'11"	2.70 m / 8'10"	16.00 m	32.00 m / 105'0"
Civic	6 m / 19'8"	7.23 m / 23'9"	3.60 m / 11'10"	16.83 m	33.66 m / 110'5"

WIDEST BUILDING WIDTH

33.66 m
110'5" · Civic

NARROWEST BUILDING WIDTH

14.73 m
48'4" · Private Office

SPREAD — WIDEST MINUS NARROWEST

18.93 m
62.1 ft

The ±5 ft precision question. Tiles must compose to a single floor-plate width across all use types on a floor. tool-buildingwidth resolves this by computing each floor's tenant-mix maximum width at design time, then redistributing Zone 2 or constraining the mix.

The Floor Plate

Tiles are the assembly units — Small Tile = 2,700 SF, Large Tile = 4,900 SF — and they compose to fill the plate exactly. Each Tile is an independent climate zone: a tenant who wants thermostat control leases a Tile. Three example compositions below.

Composition A — Corporate-heavy

Three large corporate tiles (Tile F, 4,900 SF each) plus two standard corporate tiles (Tile A, 2,700 SF each). Five independent climate zones — each tenant controls their own thermostat. Total: 20,100 SF.

Tile F 4,900 SF 1 climate zone	Tile F 4,900 SF 1 climate zone	Tile F 4,900 SF 1 climate zone	Tile A 2,700 SF 1 climate zone	Tile A 2,700 SF 1 climate zone
Tiles: 5		Climate zones: 5		Total: 20,100 SF (target 20,000)

Composition B — Private Office heavy

Maximum climate-zone count. Two Tile G (4,900 SF, 10 climate zones each) plus four Tile B-1 (2,700 SF, 5 climate zones each) end caps. Twenty private licensed offices and twenty more in end caps. Forty independent thermostats on one floor.

Tile B-1 2,700 SF 5 climate zones	Tile G 4,900 SF 10 climate zones	Tile B-1 2,700 SF 5 climate zones	Tile B-1 2,700 SF 5 climate zones	Tile G 4,900 SF 10 climate zones	Tile B-1 2,700 SF 5 climate zones
Tiles: 6		Climate zones: 40		Total: 20,600 SF (target 20,000)	

Composition C — Mixed Professional Centre

Two Tile H (large professional, 3 climate zones each), one Tile C-2 (Medium professional pair), one Tile C-1 (Medium + Small), and two end-cap mixed tiles (E-1 / E-2). Tenants: Medical anchor + Business cluster + Private Office end caps.

Tile E-1 2,700 SF 3 climate zones	Tile H 4,900 SF 3 climate zones	Tile C-2 2,700 SF 2 climate zones	Tile H 4,900 SF 3 climate zones	Tile C-1 2,700 SF 2 climate zones	Tile E-2 2,100 SF 3 climate zones
Tiles: 6		Climate zones: 16		Total: 20,000 SF (target 20,000)	

Tile = Climate Zone. Composition B (Private Office heavy) yields forty independent thermostats on one floor. Source token: `bim.tile.design-principles.climate-zone-autonomy`.

What Comes Next

This HTML is the specification document for `tool-buildingwidth`, a Rust engine that reads the DTCG tokens listed at right and computes floor plate dimensions, tile compositions, and demising-wall coordinates from a `tenant-mix` input.

The engine outputs IFC-compatible geometry and a leasing-plan summary; this deck is what the engine will render as a human preview.

Every metre, every square foot, every percentage on these slides is read live from the DTCG token files at page load. When the tokens change — a new key plan sample, a corridor width update — the deck changes with them. The engine, when shipped, shares the same source.

Source: `woodfine-design-bim` DTCG token library v0.0.2

SOURCE TOKEN FILES

building-width-calculator.dtcg.json
floor-plate-standards.dtcg.json
professional-office-subtypes.dtcg.json
tile-system.dtcg.json

FUTURE CONSUMER

pointsav-monorepo/
 crates/tool-buildingwidth/